

Question Paper: Machine Learning

****Machine Learning Question Paper****

****Instructions:****

- The question paper consists of three sections: Multiple Choice Questions, Short Questions, and Long/Descriptive Questions.
- All questions are compulsory.
- Use a pencil to fill in the answers for Multiple Choice Questions.
- Use a pen to answer Short and Long/Descriptive Questions.
- Write your name and roll number clearly at the top of the answer sheet.

****Section A: Multiple Choice Questions (1-10)****

1. Which of the following is a type of machine learning algorithm?

- A) Linear Regression
- B) Web Development
- C) Data Mining
- D) Computer Networks

Answer: A) Linear Regression

2. What is the primary goal of a neural network in machine learning?

- A) To classify data
- B) To predict continuous values

Question Paper: Machine Learning

C) To cluster similar data points

D) All of the above

Answer: D) All of the above

3. Which of the following techniques is used to avoid overfitting in machine learning models?

A) Regularization

B) Normalization

C) Feature Scaling

D) Cross Validation

Answer: A) Regularization

4. What is the term for the process of training a machine learning model on a large dataset?

A) Supervised Learning

B) Unsupervised Learning

C) Reinforcement Learning

D) Model Deployment

Answer: A) Supervised Learning

5. Which of the following is a popular library used for machine learning in Python?

A) NumPy

B) Pandas

Question Paper: Machine Learning

C) Scikit-Learn

D) TensorFlow

Answer: C) Scikit-Learn

6. What is the term for the measure of how well a machine learning model performs on unseen data?

A) Accuracy

B) Precision

C) Recall

D) Generalization

Answer: D) Generalization

7. Which of the following algorithms is used for clustering in machine learning?

A) K-Means

B) K-Nearest Neighbors

C) Support Vector Machines

D) Decision Trees

Answer: A) K-Means

8. What is the term for the process of selecting the most relevant features from a dataset?

A) Feature Extraction

Question Paper: Machine Learning

- B) Feature Selection
- C) Dimensionality Reduction
- D) Data Preprocessing

Answer: B) Feature Selection

9. Which of the following is a type of deep learning model?

- A) Convolutional Neural Network
- B) Recurrent Neural Network
- C) Autoencoder
- D) All of the above

Answer: D) All of the above

10. What is the term for the process of using machine learning models to make predictions on new, unseen data?

- A) Training
- B) Testing
- C) Deployment
- D) Inference

Answer: D) Inference

****Section B: Short Questions (11-15)****

Question Paper: Machine Learning

11. What is machine learning, and how does it differ from traditional programming? (3 marks)
12. Explain the concept of supervised learning and provide an example of a supervised learning algorithm. (4 marks)
13. What is the purpose of data preprocessing in machine learning, and what are some common techniques used? (4 marks)
14. Describe the difference between a neural network and a decision tree, and provide an example of when each would be used. (4 marks)
15. What is overfitting in machine learning, and how can it be prevented? (3 marks)

****Section C: Long/Descriptive Questions (16-18)****

16. Describe the process of training a machine learning model, including data preparation, model selection, and hyperparameter tuning. Provide an example of a machine learning model and its application. (8 marks)
17. Explain the concept of deep learning and its applications in computer vision and natural language processing. Provide an example of a deep learning model and its architecture. (9 marks)
18. Discuss the importance of model evaluation in machine learning, including metrics such as

Question Paper: Machine Learning

accuracy, precision, and recall. Provide an example of how to evaluate a machine learning model using these metrics. (10 marks)

****Answer Key:****

****Section A: Multiple Choice Questions****

1. A) Linear Regression
2. D) All of the above
3. A) Regularization
4. A) Supervised Learning
5. C) Scikit-Learn
6. D) Generalization
7. A) K-Means
8. B) Feature Selection
9. D) All of the above
10. D) Inference

****Section B: Short Questions****

11. Answer should include a brief definition of machine learning and its difference from traditional programming.
12. Answer should include a definition of supervised learning and an example of a supervised learning algorithm, such as linear regression.

Question Paper: Machine Learning

13. Answer should include a brief explanation of the purpose of data preprocessing and some common techniques, such as normalization and feature scaling.

14. Answer should include a description of the difference between a neural network and a decision tree, and an example of when each would be used.

15. Answer should include a brief definition of overfitting and some methods for preventing it, such as regularization and cross-validation.

****Section C: Long/Descriptive Questions****

16. Answer should include a description of the process of training a machine learning model, including data preparation, model selection, and hyperparameter tuning. An example of a machine learning model and its application should be provided.

17. Answer should include a description of the concept of deep learning and its applications in computer vision and natural language processing. An example of a deep learning model and its architecture should be provided.

18. Answer should include a discussion of the importance of model evaluation in machine learning, including metrics such as accuracy, precision, and recall. An example of how to evaluate a machine learning model using these metrics should be provided.